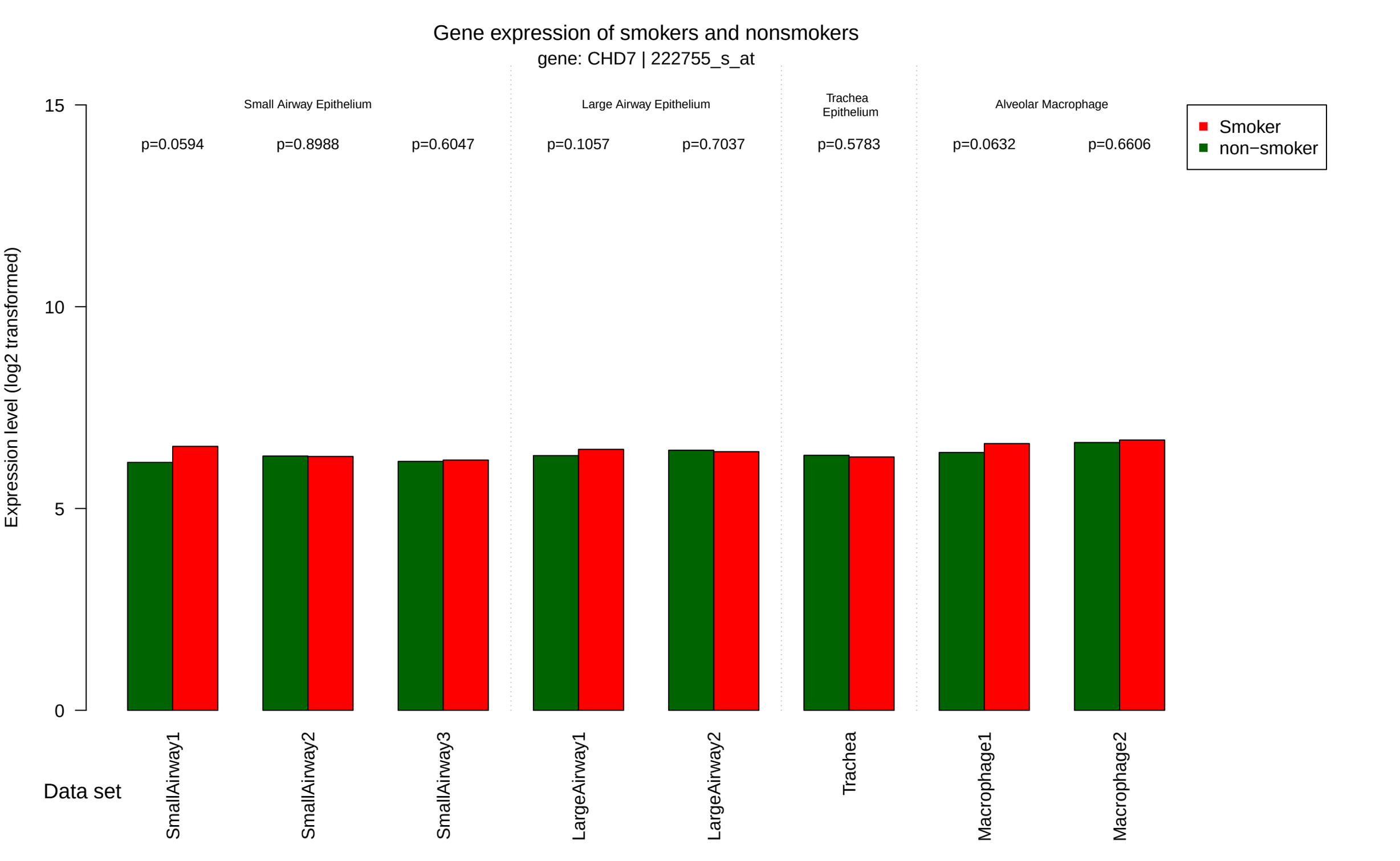
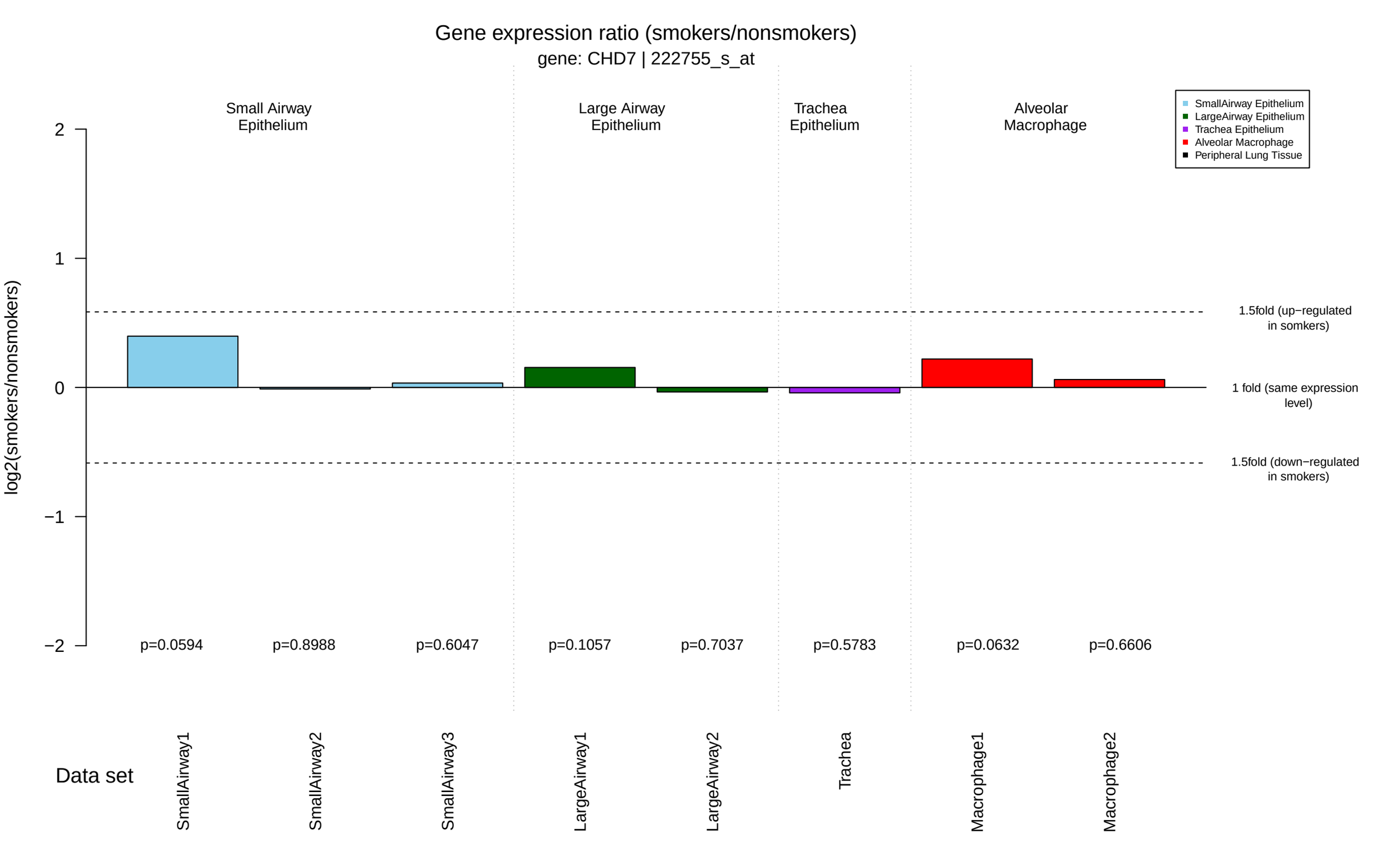
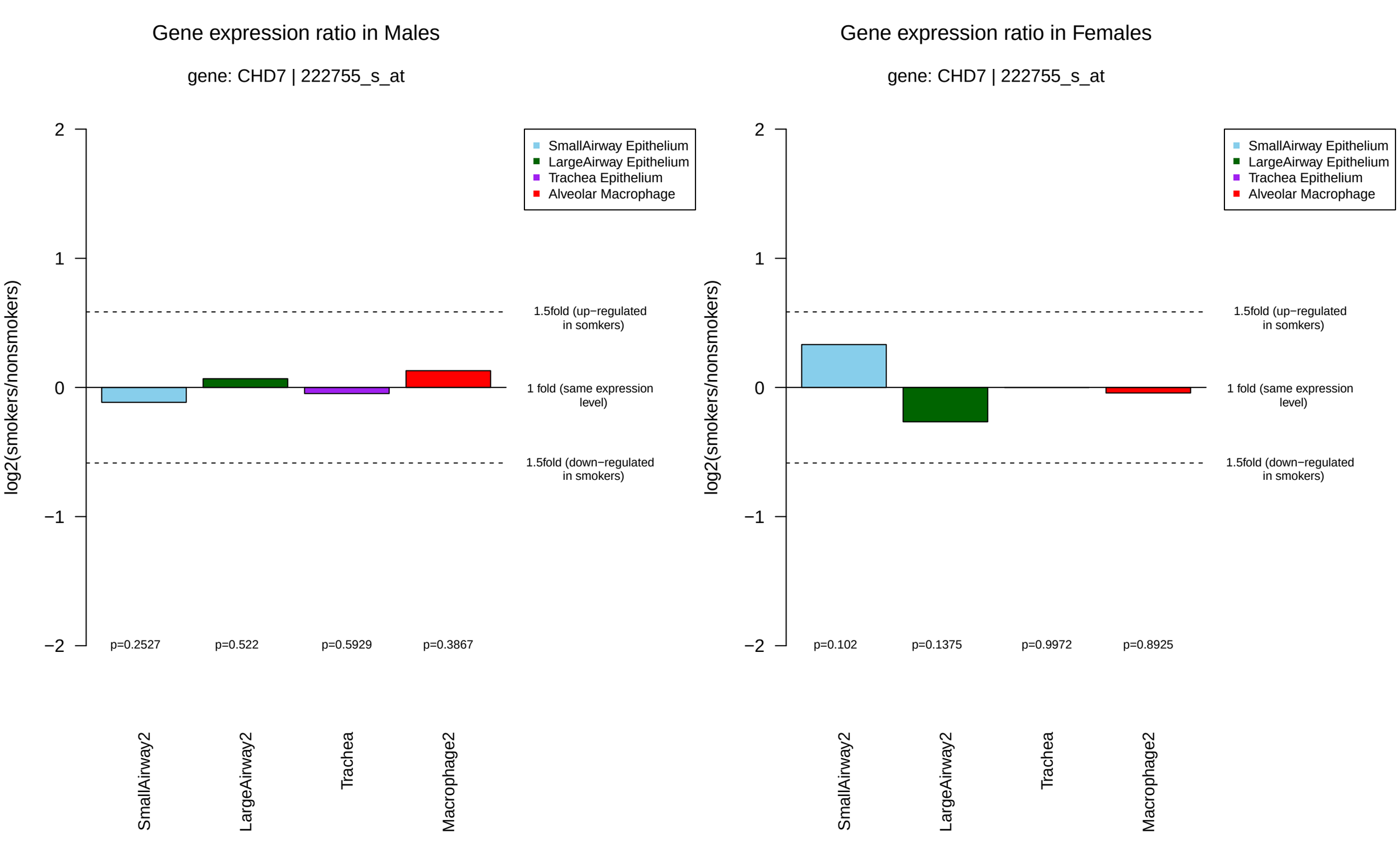
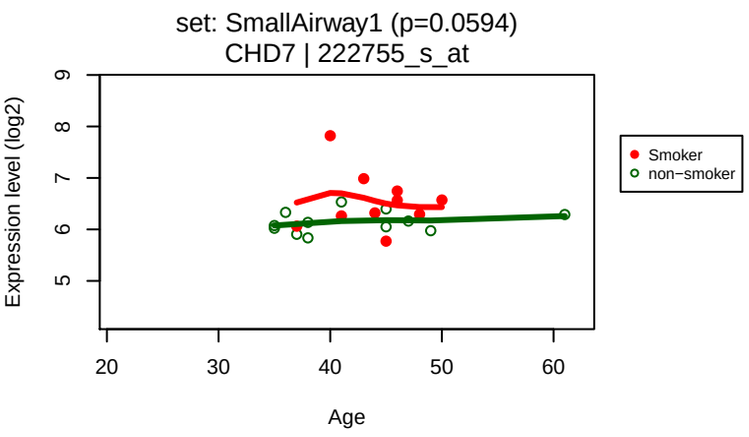
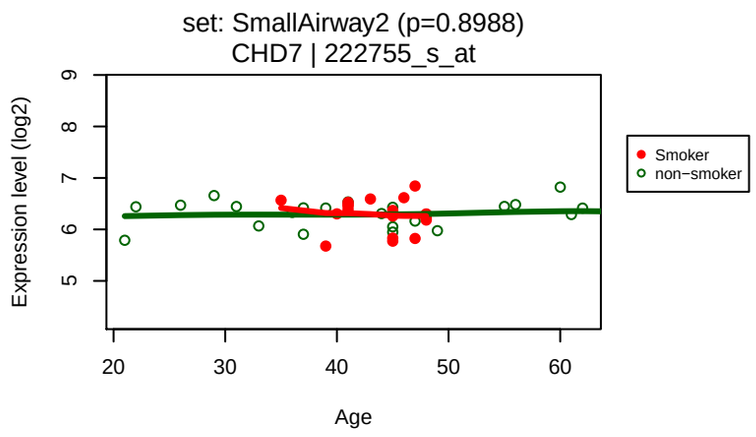
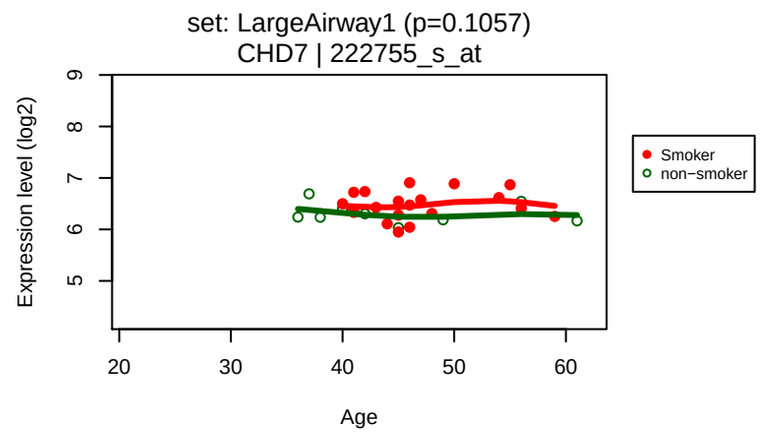

Gene expression of smokers and nonsmokers
gene: CHD7 | 222755_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0594
p=0.8988
p=0.6047
p=0.1057
p=0.7037
p=0.5783
p=0.0632
p=0.6606
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

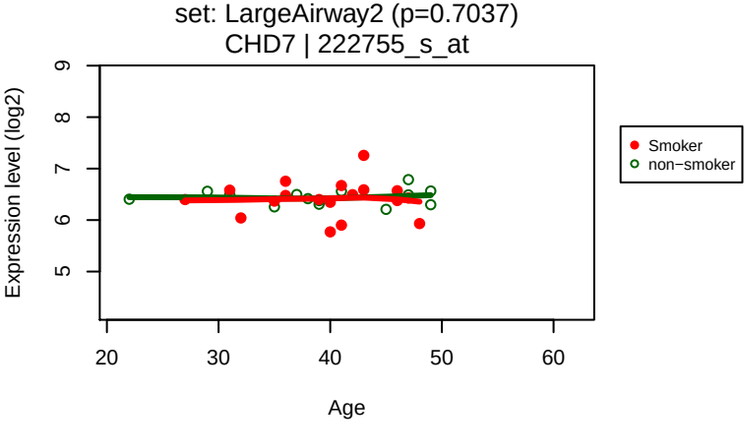

Gene expression ratio (smokers/nonsmokers)
gene: CHD7 | 222755_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0594
p=0.8988
p=0.6047
p=0.1057
p=0.7037
p=0.5783
p=0.0632
p=0.6606
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

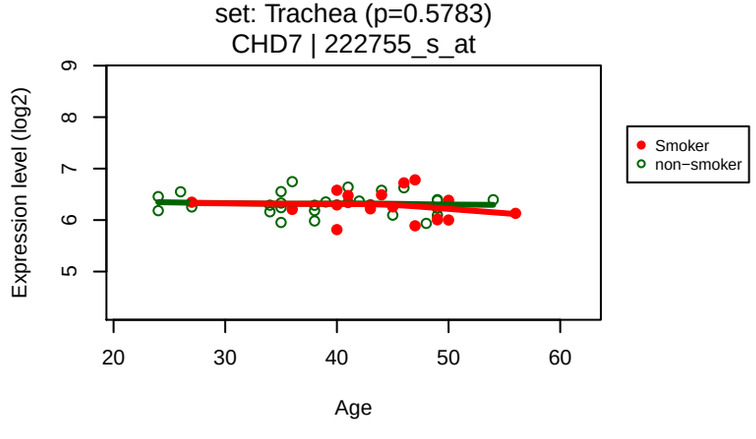

Gene expression ratio in Males
gene: CHD7 | 222755_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2527
p=0.522
p=0.5929
p=0.3867
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHD7 | 222755_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.102
p=0.1375
p=0.9972
p=0.8925
SmallAirway2
LargeAirway2
Trachea
Macrophage2

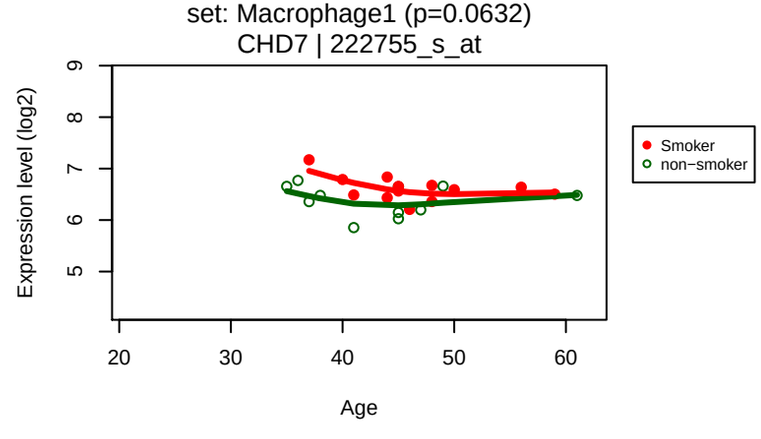

set: SmallAirway1 (p=0.0594)
CHD7 | 222755_s_at
Expression level (log2)
Age
Smoker
non−smoker

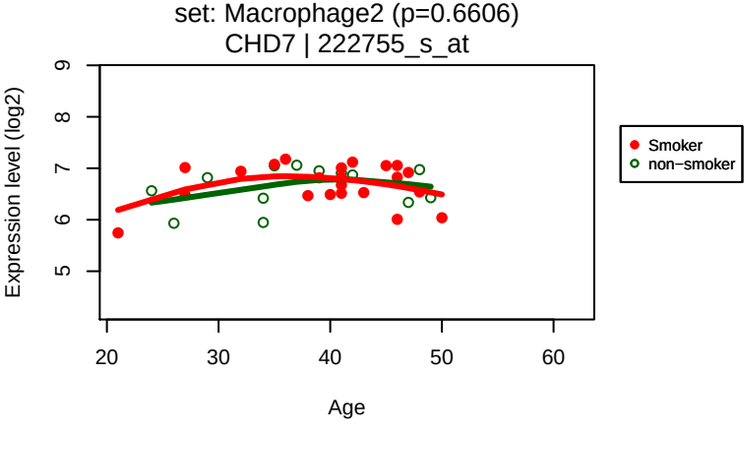

set: SmallAirway2 (p=0.8988)
CHD7 | 222755_s_at
Expression level (log2)
Age
Smoker
non−smoker

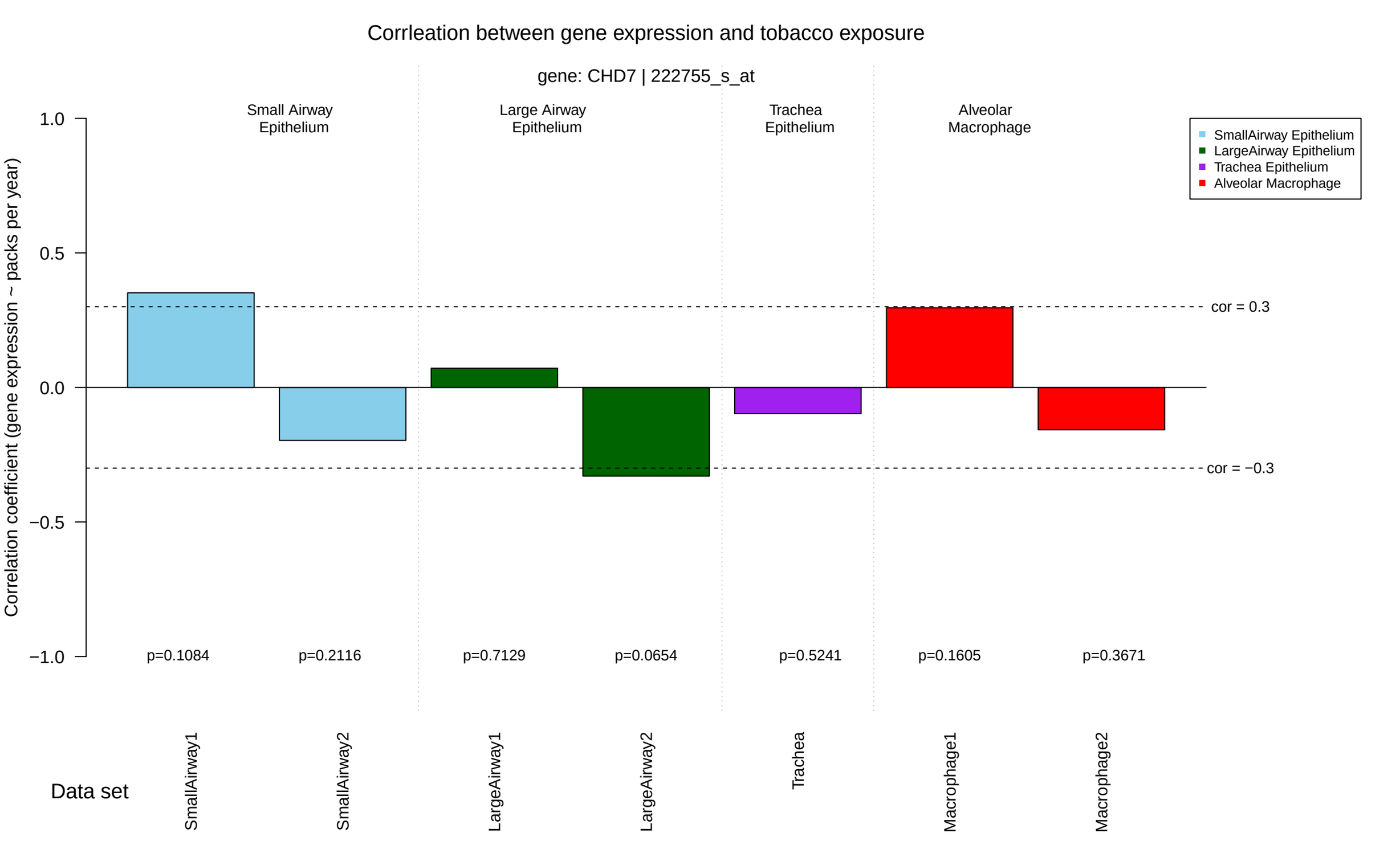

set: LargeAirway1 (p=0.1057)
CHD7 | 222755_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7037)
CHD7 | 222755_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5783)
CHD7 | 222755_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0632)
CHD7 | 222755_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6606)
CHD7 | 222755_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHD7 | 222755_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1084
p=0.2116
p=0.7129
p=0.0654
p=0.5241
p=0.1605
p=0.3671
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2